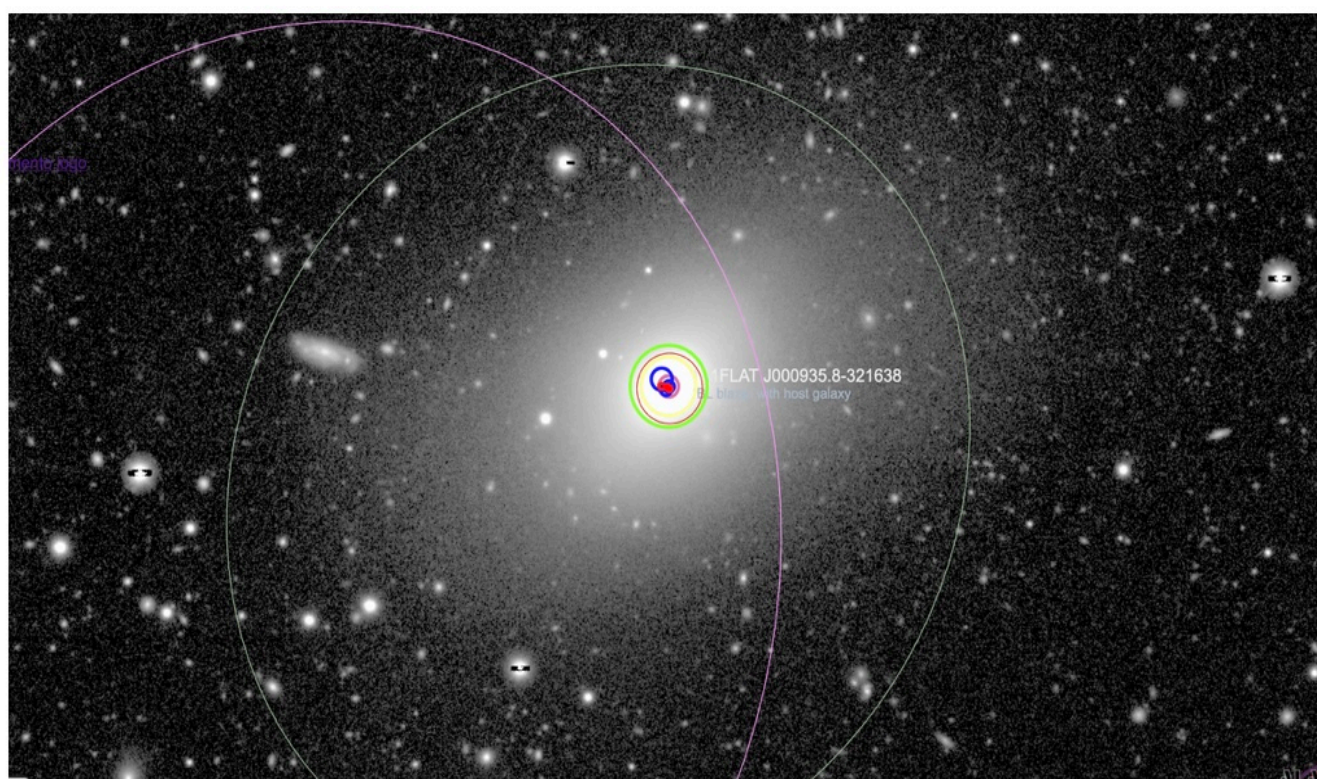


The 2FLAT project

An update of the 1FLAT sample based on the recently published FL16Y catalog, with the addition of redshifts



P. Giommi, M. Doro

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1 Status of the activity and preliminary results

A few months ago a new Fermi-LAT catalog, called FL16Y and based on 16 years of data taking, has been published (Ballet et al. 2026). It includes 7,224 sources, 5,137 of which are located above the Galactic plane ($|b| > 10^\circ$).

After the successful work on 1FLAT, we have an opportunity to consolidate our reputation and likely lead the field of Fermi-LAT source association by publishing and promoting a follow-up catalog (2FLAT, based on FL16Y) within a short time. I have been working hard to improve the ERCI tool to make it more efficient, particularly at finding counterparts that lack X-ray detection, since a fraction of these counterparts were not detected during the 1FLAT work. These sources are detectable with the new version of ERCI (V2.31+), which is available within Firmamento.

Using ERCI V2.31 I checked all the FL16Y sources that are not in 4FGL-DR4, and also all sources that are in both catalogs and have no counterpart in FL16Y and in 1FLAT.

The preliminary results (updated late-June 2026) are very encouraging,

- the current version of the 2FLAT sample includes a total of 4,358 blazars associated with as many FL16Y $|b| > 10^\circ$ sources. This is **an increase of over 550 associations compared with 1FLAT**.
- 186 FL16Y ($|b| > 10^\circ$) sources are associated with non-AGN sources (Pulsars, sources in the Magellanic clouds and other nearby galaxies)
- about **550 γ -ray sources that have no association in the FL16Y catalog are associated with 2FLAT blazars**.
- **the percentage of FL16Y γ -ray sources above the Galactic plane ($|b| > 10$) that remains unassociated reduces to $\sim 11.2\%$.**

The significant number of new associations is likely due to:

- a) Additional 600 sources (with many new pulsars) in the FL16Y catalog ($|b| > 10^\circ$) compared to 4FGL-DR4.
- b) Updated FL16Y γ -ray error regions that better include counterparts, like in the case illustrated in Fig. 1.
- c) Additional multi-frequency catalogs that have been recently added to ERCI.
- d) The improved search and identification algorithm used in the new version of the ERCI tool.

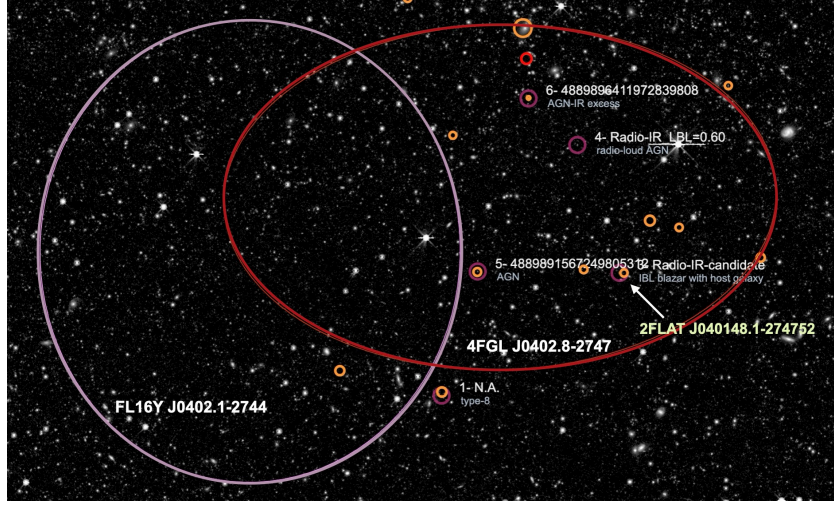


Figure 1: TBW

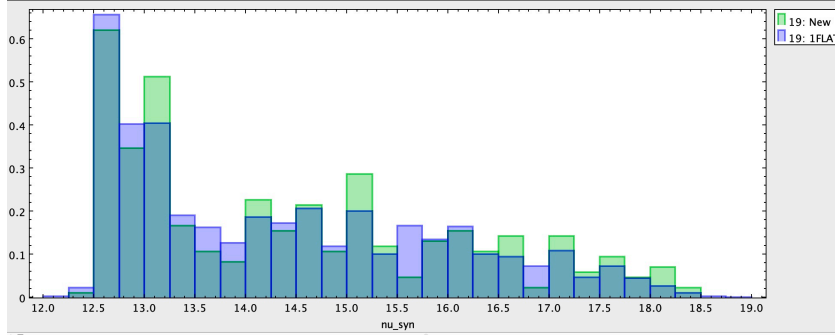


Figure 2: Normalized histograms of ν_{peak} values of 1FLAT blazars and newly associated sources: statistically identical.

2 What to do next

All the co-authors of the 1FLAT paper are invited to participate in the preparation of the 2FLAT paper and in future activities . Additional people can also participate if interested.

The encouraging preliminary results described in the previous section are the outcome of a first run through the data and need to be confirmed. In addition, it is necessary to a) verify the correctness of all the 1FLAT published sources to make sure that they are still the best counterpart of FL16Y sources and b) correct possible problems in the 1FLAT paper: for instance in a few cases the positions could be slightly offset, or the ν_{peak} and $\nu_{F_{\nu}}$ values could be a bit different with the new multi-frequency data that is available now.

Initial list of **things to do**

- Check the new associations (sources with names starting with 2FLAT in the third column of the table shown in fig. 3) to secure a second opinion on new identifications.

- Check the old 1FLAT sources to correct possible problems and verify that these sources are consistent with the FL16Y error ellipses.
- Add redshifts, using the "get_redshift" tool in Firmamento, which now includes the 1CGH catalog featuring accurate redshifts of γ -ray detected blazars
- Write the 2FLAT paper.

Possible timeline:

- 1) **Beginning of summer:** complete the checks and publish only on ArXiv the 2FLAT preliminary catalog (The Fermi team does that before publishing the formal paper)
- 2) **After summer:** present the 2FLAT catalog at conferences. The first one could be the Heidelberg γ -ray meeting, another one could be the Workshop in honor of Paolo Padovani at ESO.
- 3) **One year timescale:** complete the paper and submit to ApJS.

3 Project work description

For people who wish to co-author the paper.

To actively work on the project you should use the shared google sheet linked to the figure below (Fig.3). Click on the figure to open the sheet.

1	16YName	4FGLName	1FLATName	16Y_ASSOC1	New/confirmed	WHO-1	Second evaluat WHO-2	comment	nu_syn	nuFnu_sy
75	FL16Y_J0015.9+2441	4FGL_J0015.9+2440	1FLAT_J001803.6+244014	G86 J0016+2440	Confirmed 1FLAT	MG			14.5	-11.8
76	FL16Y_J0016.2+0017	4FGL_J0016.2+0016	1FLAT_J001611.1+001513	S3 0013-00	Confirmed 1FLAT	MG			12.5	-12.2
77	FL16Y_J0017.0+0651	4FGL_J0017.0+0649	1FLAT_J001709.4+065033	PMN J0017-0650	Confirmed 1FLAT	MG			14.2	-11.9
78	FL16Y_J0017.0+4606	4FGL_J0017.1+4605			No counterpart	P.G.				
79	FL16Y_J0017.0+1656	4FGL_J0016.5+1702	1FLAT_J001540.0+170041		Remove 1FLAT	MG				
80	FL16Y_J0017.5+0512	4FGL_J0017.5+0514	1FLAT_J001735.9+051242	PMN J0017-0512	Confirmed 1FLAT	MG		the 1FLAT association is not four	12.7	-12.6
81	FL16Y_J0017.7+1451	4FGL_J0017.8+1455	1FLAT_J001736.9+145101	G86 J0017+1450	Confirmed 1FLAT	MG			13.3	-11.5
82	FL16Y_J0018.5+2947	4FGL_J0018.4+2946	1FLAT_J001827.8+294730	RBS 0042	Confirmed 1FLAT	MG			14.4	-11.8
83	FL16Y_J0018.9+5111	4FGL_J0018.8+5112			No counterpart	P.G.			17.4	-11.6
84	FL16Y_J0019.2+3042		2FLAT_J001942.7+303119	PKS 0017-307	Confirmed F16Y	P.G.		LBL		
85	FL16Y_J0019.3+5641	4FGL_J0019.2+5640	1FLAT_J001926.5+564142	PMN J0019-5641	Confirmed 1FLAT	MG			13.3	-12.6
86	FL16Y_J0019.3+8152	4FGL_J0019.3+8152	1FLAT_J001921.1+815250	PMN J0019-8152	Confirmed 1FLAT	P.G.			14.7	-11.4
87	FL16Y_J0019.4+2606	4FGL_J0018.8+2611	1FLAT_J001939.8+260253	4C +25.01	Confirmed 1FLAT	P.G.			12.5	-11.2
88	FL16Y_J0019.6+2020	4FGL_J0019.6+2022	1FLAT_J001937.8+202145	PKS 0017+200	Confirmed 1FLAT	P.G.			12.8	-12.1
89	FL16Y_J0019.6+7327	4FGL_J0019.6+7327	1FLAT_J001945.8+732730	SS 0016+73	Confirmed 1FLAT	P.G.			12.5	-11.8
90	FL16Y_J0020.9+0322	4FGL_J0021.0+0322	1FLAT_J002050.2+032357	2MASS J00205023+0323578	Confirmed 1FLAT	P.G.			15	-12.4
91	FL16Y_J0021.3+5917	4FGL_J0022.0+5921	1FLAT_J002127.3+591948	1eRASS J002127.3-591945	Confirmed 1FLAT	P.G.			14.7	-12.5
92	FL16Y_J0021.5+2551	4FGL_J0021.5+2552	1FLAT_J002132.5+255049	CRATES J002132.55-255049.3	Confirmed 1FLAT	P.G.			14.4	-11.7
93	FL16Y_J0021.6+2220	4FGL_J0021.5+2221	1FLAT_J002201.5+222103	PMN J0022-2221	Confirmed 1FLAT	P.G.			12.8	-12.9
94	FL16Y_J0021.7+0859	4FGL_J0021.6+0855	1FLAT_J002142.2+090044	NVSS J002142-090044	Confirmed 1FLAT	P.G.			14.3	-12.4
95	FL16Y_J0021.7+4531	4FGL_J0021.7+4533	2FLAT_J002159.7+453008	B3 0019+451	Confirmed F16Y	P.G.			13.2	-12.6
96	FL16Y_J0022.0+5140	4FGL_J0021.9+5140	1FLAT_J002200.1+514024	1RXS J002159.2+514028	Confirmed 1FLAT	P.G.			15.4	-11.5

Figure 3: The FL16Y/2FLAT google sheet to be used as a working sheet

Table description:

First column: name of the γ -ray source in the FL16Y catalog

Second column: name of the 4FGL source corresponding to the FL16Y source of the first column. An empty value implies that the FL16Y source is new and was not in 4FGL.

Third column: The 1FLAT name in case this source was identified in the 1FLAT paper. Alternatively the 2FLAT name in case of a new association.

Fourth column: name of the association of the FL16Y source in the FL16Y catalog.

Fifth column: Flag to be used to describe your decision on the identification of the γ -ray source, after the inspection. The options are shown in fig 4.

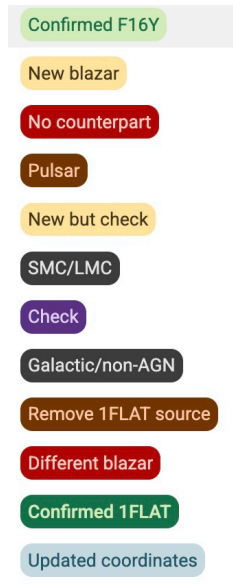


Figure 4: The list of options to be used to describe our decision

The most important flags are:

Confirm F16Y: used when the FL16Y source is new (it was not in the 4FGL catalog) and we confirm the association of the FL16Y catalog reported in column 4. In this case column 3 should list the 2FLAT name.

New blazar: our inspection finds that the γ -ray source can be associated with a blazar that is NOT in 1FLAT and is NOT already associated in FL16Y (that is column 4 is empty or gives a source different from our identification)

Confirmed 1FLAT: when our check confirms that the 1FLAT source is the correct counterpart of the FL16Y source.

Updated coordinates: to be used when we find that the coordinates of the 1FLAT source are not precise and need to be updated. The meaning of the other flags are obvious.

Sixth column The initials of the person who performed the check.

4 List of tasks

4.1 Task 1: Verify the correctness of 1FLAT sources

Task description

Verify that 1FLAT catalog sources are the correct counterparts to the FL16Y γ -ray sources and are ready for inclusion in the 2FLAT catalog.

Step-by-step procedure

1. **Select a FL16Y source** where the 1FLAT source needs to be checked. That is a source for which column E "**check/confirmed**" is blank (not set to anything).
2. **Click on the name of the FL16Y source (first column)**. This will open a Firmamento page showing the result of the ERCI tool and HIPS images. The ERCI tool should find the counterpart
3. Check that our 1FLAT/ERCI source is ok. In most cases it coincides with the 4FGL/4LAC (yellow circle) or the FL16Y (green circle) counterpart.
4. Click on the name of the 1FLAT source (third column). This will open a Firmamento page showing the SED (with the BLAST results below) and HIPS images.
5. **Check that the SED is ok**: the radio to X-ray data must be consistent with the γ -ray data (in particular FL16Y data) and the overall SED is typical of blazars.
 - If yes (most of the cases) then set the value of Column E to "Confirmed 1FLAT".
 - If the radio to γ -ray SED is inconsistent with that of a blazar (hopefully a rare case) a) set column E to "Remove 1FLAT", b) change the color of the row to light red, c) write a short text in the "comment" column, and c) report this event at the next video meeting.
6. **Check the position of the 1FLAT source in the image**. If the position of the optical source is within one or two arc-seconds from the 1FLAT source don't do anything. If the position is instead some arc-seconds away update the Ra and Dec coordinates in columns Q and R ("1/2FLATra and 1/2FLATdec). After that set column E to "Updated coordinates".
7. sign selecting your initials in column F.

You tube video

4.2 Task 2: Verify the correctness of a newly identified counterpart

Task description

check that blazars newly associated with FL16Y sources (selected by the FL16Y team or by PG) are OK and can be included in the 2FLAT catalog. These are sources with names starting with 2FLATJ in column "1/2FlatName".

Step-by-step procedure

1. **Select a source flagged as "Confirmed FL16Y" (light green) or "New blazar" (yellow) in column "New/confirmed"**
 - **Confirmed FL16Y:** the FL16Y catalog association listed in column "16Y_ASSOC1" is valid and confirmed by us.
 - **New blazar:** we identify a valid association and the 16Y_ASSOC1 field is blank or gives a source different from our counterpart.
2. **Check the 2FLAT association.** The name of a new association (2FLATJxxx) is listed in column C (1/2FlatName) and the corresponding precise coordinates are in columns "1/2FLATra" and "1/2FLAT-dec".
 - Click on the FL16Y source in column A (16YName)
 - check that the ERCI tools finds the 2FLAT source
 - to check the validity of the 2FLAT source follow the same instructions for 1FLAT sources
 - sign selecting your initials in column F.

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4.3 Task 3: Get a reliable value of the redshift

Task description

For each 2FLAT source we would like to publish a reliable value of its redshift. This is not a straightforward task since often widely different and unreliable values are present in the literature.

Step-by-step procedure

- In the firmamento page that shows the SED of a 1/2FLAT source click on the button "Get Redshift". This will trigger a search for redshift value in several catalogs. It often happens that the results are different and not consistent. If the SDSS or DESI-DR1 give a "spectroscopic" value check the optical spectrum, by clicking on the tab "More" in the Firmamento page and then on "SDSS-DR19" or "DESI-DR1". If there is no consensus among the catalogs (different values are give) don't do anything.
- After checking the redshift of a source write the outcome in the dedicated columns (Col J, K and L) using the predefined values (spectroscopic, photometric, featureless, etc).
- sign selecting your initials in column M.

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5 Quick Firmamento tutorial

This section briefly explains the content of the Aladin HiPS sky maps generated by Firmamento-ERCI. A more general Firmamento tutorial (still experimental and under construction) based on Fbook can be started by clicking on the icon below.



< Firmamento Tutorial

5.1 Understanding the positional errors of multi-frequency sources and of FERMI-LAT counterpart positions in ERCI

One of the outputs of the ERCI tool is an Aladin sky map displaying HiPS images together with the positions of the sources identified by the tool. The map also shows the localization error circles or ellipses for radio, IR, optical, X-ray, and γ -ray sources from a range of catalogs, as well the position of proposed counterparts of FERMI-LAT γ -ray sources from the 4FGL-DR4 and FL16Y catalogs. The different types of sources are distinguished by different colors, as shown in the legend below and illustrated in the figures that follow.

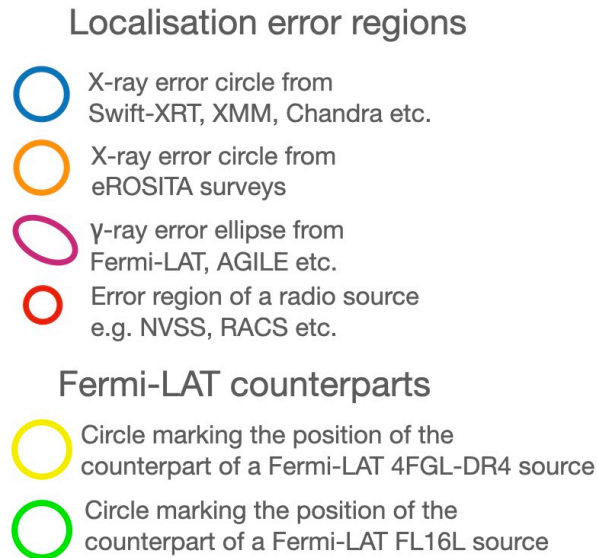


Figure 5: Legend describing the color and shape coding used in the ERCI output map.

Figure 6 shows an example of the output of the ERCI tool applied to a Fermi-LAT source listed in both the 4FGL-DR4 and FL16Y catalogs.

Figure 7 illustrates the meaning of the colored circles representing the position of FERMI-LAT proposed candidates counterparts from the 4FGL-DR4 (yellow circle), and the FL16Y (green circle) catalogs.

Figure 8 extends the cases shown in Figure 7 showing the uncertainty area of one X-ray source (blue circle representing data from e.g. Swift-XRT, XMM etc.) and one radio source (small red circle).

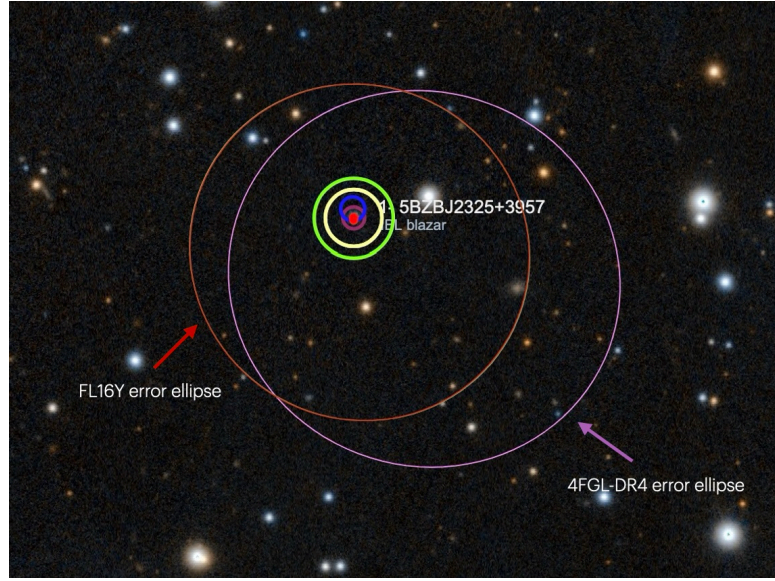


Figure 6: Example of an image generated by ERCI showing the error ellipses from the Fermi-LAT 4FGL-DR4 and the FL16Y catalogs.

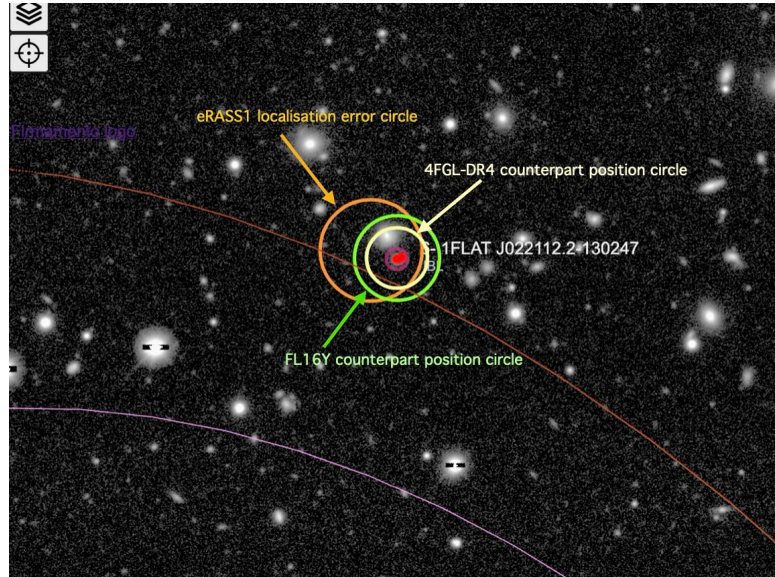


Figure 7: Detail of a ERCI-generated image showing the position of the Fermi-LAT sources and an X-ray source from the eRASS1 survey.

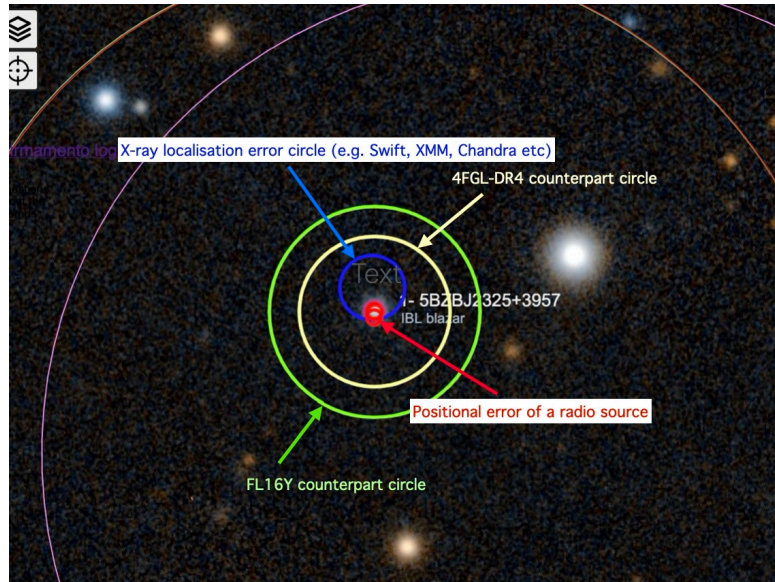


Figure 8: Detail of an ERCI-generated image showing the position of the Fermi-LAT sources as well as the error circle of a Swift-XRT X-ray source.